Before 1996, most theoretical models that incorporated extra dimensions imagined that their spatial extent was roughly Planckian (10^{-33} centimeters). As this is seventeen orders of magnitude smaller than anything resolvable using currently available equipment, without the discovery of miraculous new technology Planckian physics will remain out of reach. But if the extra dimensions are "large," meaning larger than a hundredth of a billionth of a billionth (10^{-20}) of a meter, about a millionth the size of an atomic nucleus, there is hope.

As we discussed in Chapter 13, if any of the extra dimensions are "very large"—within a few orders of magnitude of a millimeter—precision measurements of gravity's strength should reveal their existence. Such experiments have been under way for a few years and the techniques are being rapidly refined. So far, no deviations from the inverse square law characteristic of three space dimensions have been found, so researchers are pressing on to smaller distances. A positive signal would, to say the least, rock the foundations of physics. It would provide compelling evidence of extra dimensions accessible only to gravity, and that would give strong circumstantial support for the braneworld scenario of string/M-THEORY.

If the extra dimensions are large but not very large, precision gravity experiments will be unlikely to detect them, but other indirect approaches remain available. For example, we mentioned earlier that large extra dimensions would imply that gravity's intrinsic strength is greater than previously thought. The observed weakness of gravity would be attributed to its leaking out into the extra dimensions, not to its being fundamentally feeble; on short distance scales, before such leakage occurs, gravity would be strong. Among other implications, this means that the creation of tiny black holes would require far less mass and energy than it would in a universe in which gravity is intrinsically far weaker. In Chapter 13, we discussed the possibility that such microscopic black holes might be produced by high-energy proton-proton collisions at the Large Hadron Collider, the particle accelerator now under construction in Geneva, Switzerland, and slated for completion by 2007. That is an exciting prospect. But there is another tantalizing possibility that was raised by Alfred Shapere, of the University of Kentucky, and Jonathan Feng, of the University of